Ensemble-Based Spatial Snow Water Equivalent Estimates Using the SWE Evaluation Tool for the Rio Grande Basin

February 15, 2026

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Introduction

No spaceborne sensor can directly measure snow water equivalent (SWE). Therefore, the spatial distribution of SWE must be estimated from remotely sensed observations combined with modeling approaches. Existing gridded SWE products exhibit documented accuracy limitations across both mountainous and non-mountainous regions, resulting in a wide range of SWE estimates for any given location. To address this uncertainty, the New Mexico Office of the State Engineer (NM OSE), in collaboration with the Institute of Arctic and Alpine Research (INSTAAR), developed the SWE Evaluation Tool (SWEET). SWEET integrates multiple suitable SWE data products into an ensemble framework, with the objective of producing near-real-time spatial SWE estimates for the Rio Grande Basin that are, on average, as accurate as or more accurate than any individual product alone.

Figure 1 shows the near-real-time (February 15, 2026) ensemble SWE estimates for the Rio Grande Basin. Subsequent sections of this report present detailed ensemble-based SWE maps, along with summaries by subbasin and elevation band. In addition to the ensemble results highlighted in this report, figures and tables for each individual SWE product are also provided for download to support transparency and further analysis. These data sets and visualizations are available on the GitHub repository.

For ease of navigation, hyperlinks are provided throughout this report to help readers quickly access basin-specific sections and supporting data tables.

Rio Grande Basin SWE Overview

Rio Grande Headwaters

Upper Rio Grande

Rio Grande-Elephant Butte (Middle Rio Grande)

About SWEET

Rio Grande Basin SWE Overview

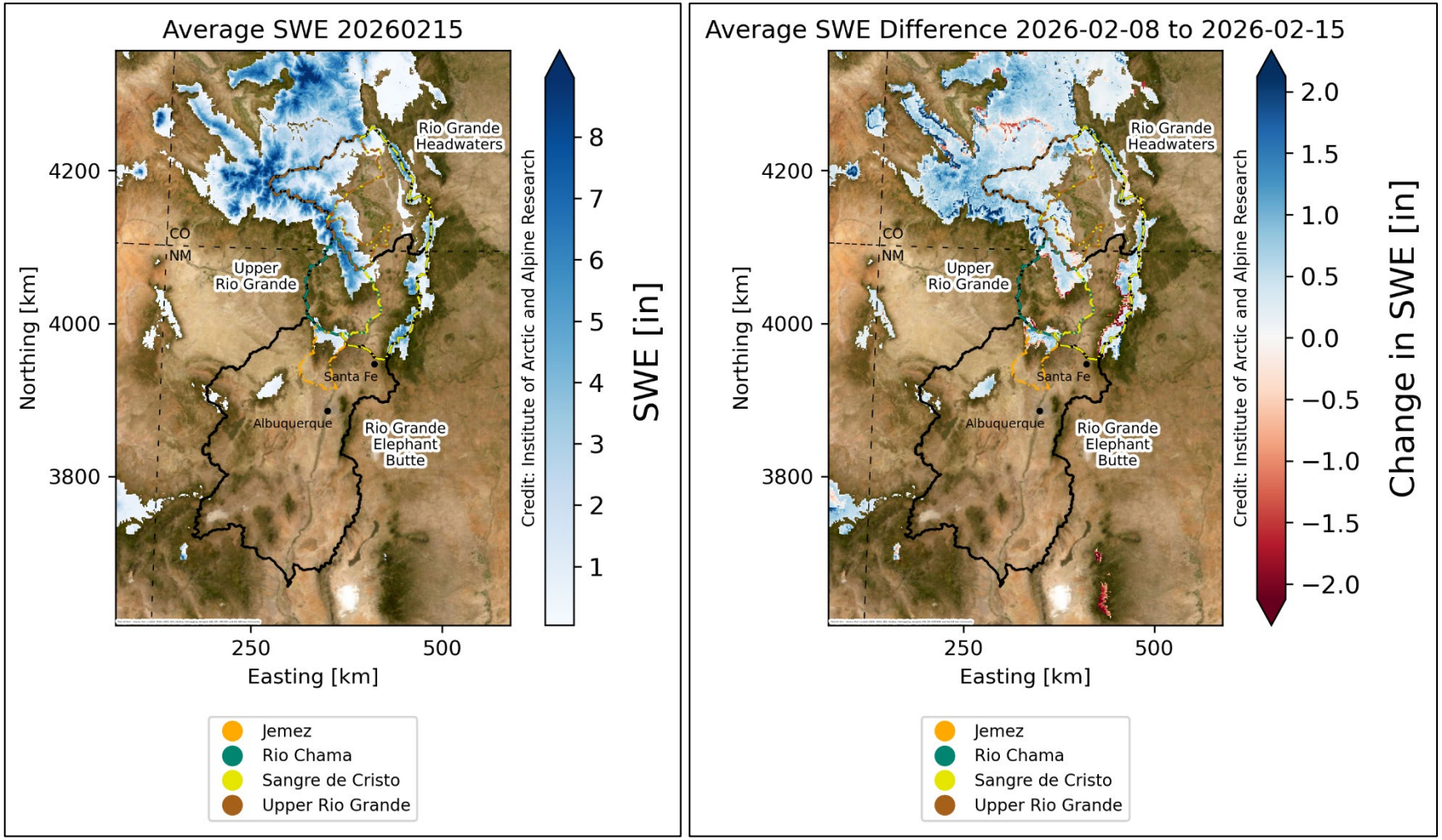

Figure 1. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande Basin for February 15, 2026. *SWE amounts (in.) across the Rio Grande Basin (left); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande Basin from February 8, 2026, to February 15, 2026 (right); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts.*

Table 1. Estimated ensemble SWE by HUC6 Basin for February 15, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 8, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value. Values are separated by HUC8 subbasin and totaled for each HUC8 basin (bold).*

HUC6 Basin	HUC8 Basin	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-08 (af)	Median Vol. Change since 2026-02-08 (af)
Rio Grande Headwaters	Rio Grande Headwaters	228,342	223,040	220,902 (UA_SWE)	241,084 (SNODAS)	24,928	17,103
	Alamosa-Trinchera	84,615	84,090	74,739 (CU_SWE)	95,016 (UA_SWE)	6,199	6,349
	San Luis	58,596	59,542	47,628 (CU_SWE)	68,617 (UA_SWE)	8,437	10,244
	Saguache	63,331	58,183	47,559 (CU_SWE)	84,252 (UA_SWE)	7,021	8,549
	Conejos	107,035	107,826	90,022 (CU_SWE)	123,257 (SNODAS)	6,975	8,600
	Total	**541,919**	**566,156**	**482,987 (CU_SWE)**	**576,613 (UA_SWE)**	**53,559**	**87,028**
Upper Rio Grande	Upper Rio Grande	102,128	110,531	42,335 (CU_SWE)	153,519 (SNODAS)	6,275	-6,278
	Rio Chama	145,135	116,329	109,693 (CU_SWE)	209,382 (SNODAS)	10,824	-9,378
	Total	**247,263**	**226,859**	**152,028 (CU_SWE)**	**362,901 (SNODAS)**	**17,098**	**-15,656**
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	2,403	1,002	819 (CU_SWE)	5,386 (SNODAS)	828	-580
	Jemez	21,439	20,370	9,831 (CU_SWE)	34,117 (SNODAS)	4,868	7,506
	Rio Grande-Albuquerque	143	165	0 (UA_SWE)	265 (SNODAS)	-40	27
	Arroyo Chico	1,764	1,501	1,211 (CU_SWE)	2,581 (UA_SWE)	397	1,475
	North Plains	1,211	109	0 (CU_SWE)	3,523 (UA_SWE)	1,211	109
	Rio San Jose	4,888	2,906	800 (CU_SWE)	10,959 (UA_SWE)	4,010	2,619
	Plains of San Agustin	265	219	46 (CU_SWE)	531 (UA_SWE)	204	191
	Rio Salado	12	2	0 (CU_SWE)	33 (SNODAS)	12	2
	Jornada Del Muerto	0	0	0	0	0	0
	Elephant Butte Reservoir	165	215	4 (UA_SWE)	277 (CU_SWE)	-23	213
	Rio Puerco	5,774	6,568	1,500 (CU_SWE)	9,253 (SNODAS)	1,411	3,571
	Total	**38,064**	**45,540**	**14,649 (CU_SWE)**	**54,003 (SNODAS)**	**12,878**	**23,339**

Table 2. Estimated ensemble SWE by specified subbasin groupings for February 15, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 8, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value. Values are separated by HUC8 subbasin and totaled for each specified subbasin grouping (bold).*

OSE Grouping	HUC8 Basin	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-08 (af)	Median Vol. Change since 2026-02-08 (af)
Upper Rio Grande	Rio Grande Headwaters	228,342	223,040	220,902 (UA_SWE)	241,084 (SNODAS)	24,928	17,103
	Saguache	63,331	58,183	47,559 (CU_SWE)	84,252 (UA_SWE)	7,021	8,549
	Conejos	107,035	107,826	90,022 (CU_SWE)	123,257 (SNODAS)	6,975	8,600
	Total	**398,708**	**412,980**	**360,620 (CU_SWE)**	**422,523 (SNODAS)**	**38,923**	**48,567**
Sangre de Cristo	San Luis	58,596	59,542	47,628 (CU_SWE)	68,617 (UA_SWE)	8,437	10,244
	Alamosa-Trinchera	84,615	84,090	74,739 (CU_SWE)	95,016 (UA_SWE)	6,199	6,349
	Upper Rio Grande	102,128	110,531	42,335 (CU_SWE)	153,519 (SNODAS)	6,275	-6,278
	Total	**245,339**	**274,164**	**164,702 (CU_SWE)**	**297,151 (SNODAS)**	**20,911**	**38,414**
Rio Chama	**Total**	**145,135**	**116,329**	**109,693 (CU_SWE)**	**209,382 (SNODAS)**	**10,824**	**-9,378**
Jemez	**Total**	**21,439**	**20,370**	**9,831 (CU_SWE)**	**34,117 (SNODAS)**	**4,868**	**7,506**

Upper Rio Grande Snowpack Update Application

The OSE/ISC Upper Rio Grande Snowpack Update Application (Application) provides water managers and users with detailed SNOTEL-based SWE data organized by the same subbasin groupings shown in Table 2. The Application is updated regularly from December through May and compiles key SWE information from SNOTEL sites within each subbasin. In addition to the Upper Rio Grande groupings (Upper Rio Grande, Sangre de Cristo, Rio Chama, and Jemez), the Application also includes the San Juan River Basin, with data summarized for the Upper San Juan, Animas, and Piedra subbasins.

The Application can be viewed here[1].

[1] Please note that access to the Application may be restricted to employees of the NM OSE/ISC at this time. We are currently working on public access.

Rio Grande Headwaters

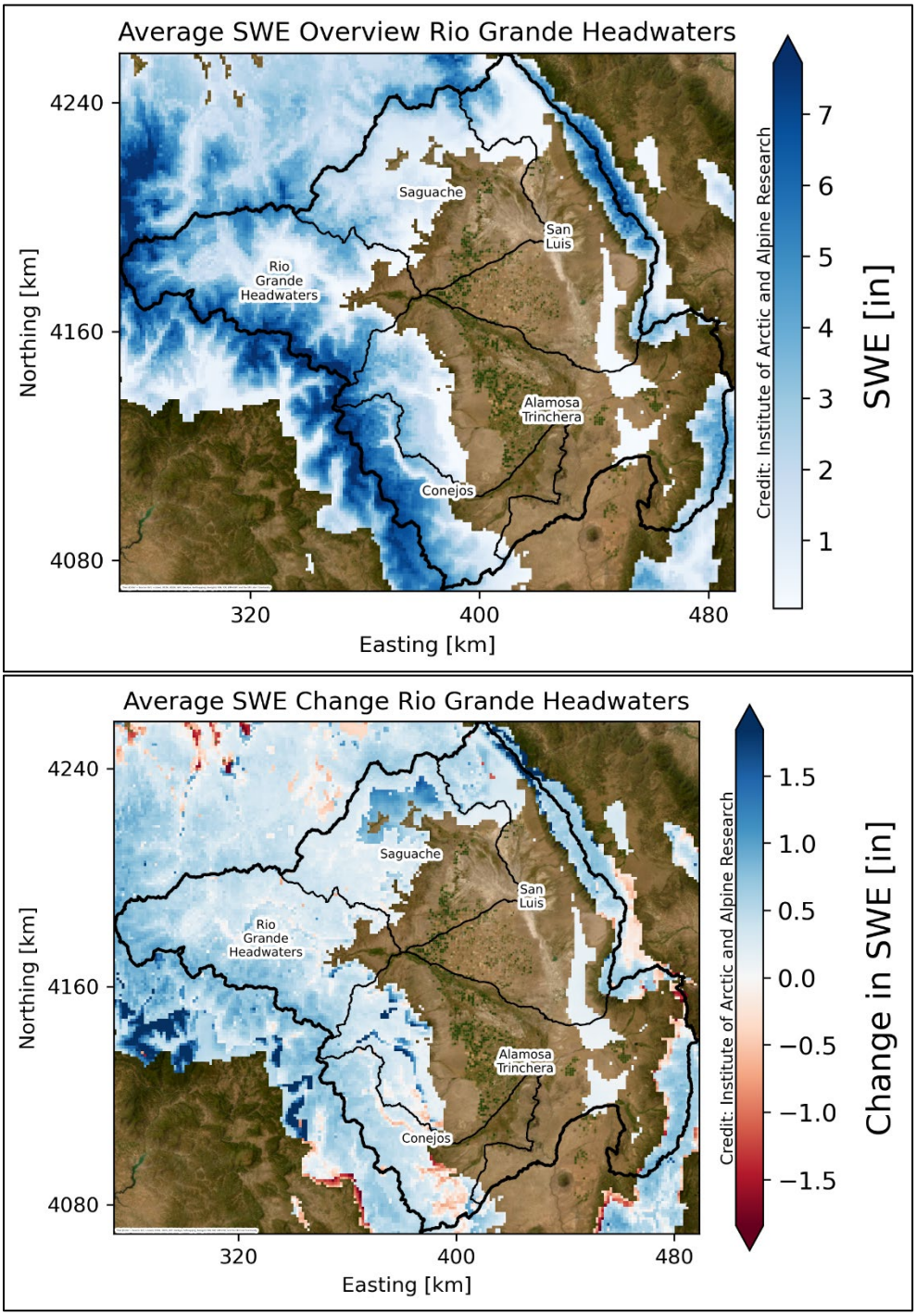

Figure 2. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande Headwaters HUC6 Basin for February 15, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Rio Grande Headwaters Basin, showing spatial patterns of SWE across subbasins including Rio Grande Headwaters, Alamosa-Trinchera, San Luis, Saguache, and Conejos (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande Headwaters from February 8, 2026, to February 15, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

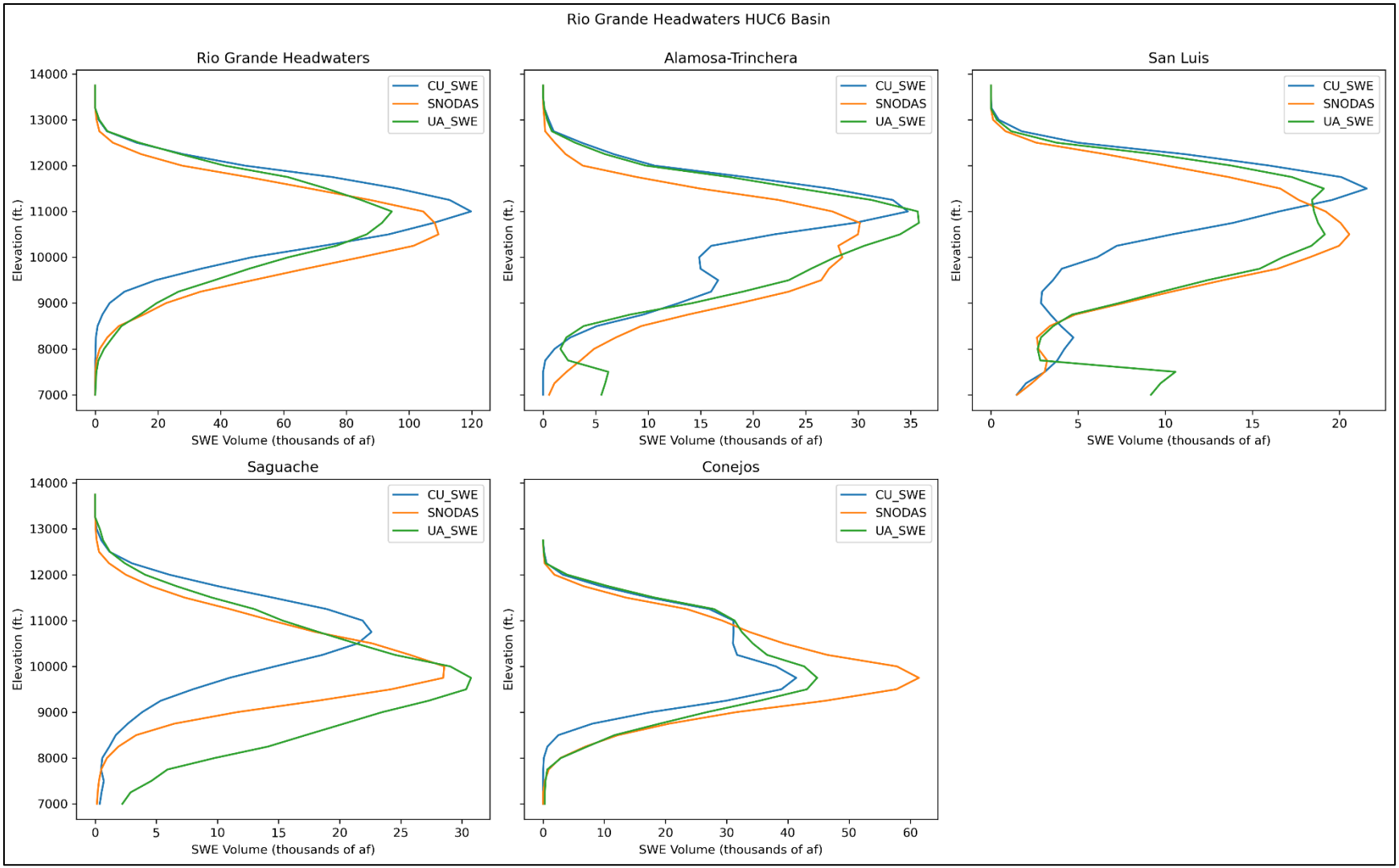

Figure 3. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Rio Grande Headwaters HUC6 Basin for February 15, 2026. *Elevation-distributed SWE volume for the Rio Grande Headwaters HUC6 Basin and associated subbasins (Rio Grande Headwaters, Alamosa–Trinchera, San Luis, Saguache, and Conejos), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 3. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Rio Grande Headwaters HUC6 Basin for February 15, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 8, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-08 (af)	Median Vol. Change since 2026-02-08 (af)
Rio Grande Headwaters	Rio Grande Headwaters	7000-8000'	3	0	0 (SNODAS, CU_SWE)	8 (UA_SWE)	3	0
		8000-9000'	1,443	1,416	100 (CU_SWE)	2,812 (UA_SWE)	355	452
		9000-10000'	15,570	19,569	4,580 (CU_SWE)	22,561 (SNODAS)	2,045	2,871
		10000-11000'	65,321	61,519	49,891 (CU_SWE)	84,554 (SNODAS)	7,678	8,473
		11000-12000'	106,246	104,493	94,508 (UA_SWE)	119,737 (CU_SWE)	10,920	14,824
		12000-13000'	38,835	41,203	27,643 (SNODAS)	47,659 (CU_SWE)	3,856	4,397
		13000-14000'	924	1,072	416 (SNODAS)	1,283 (UA_SWE)	71	40
	Alamosa-Trinchera	7000-8000'	2,043	575	0 (CU_SWE)	5,554 (UA_SWE)	-1,090	575
		8000-9000'	2,527	1,648	1,088 (CU_SWE)	4,845 (SNODAS)	164	-790
		9000-10000'	15,291	14,206	12,920 (CU_SWE)	18,748 (SNODAS)	-316	-1,847
		10000-11000'	23,719	27,801	14,865 (CU_SWE)	28,491 (SNODAS)	3,127	4,626
		11000-12000'	32,640	34,721	27,534 (SNODAS)	35,664 (UA_SWE)	3,395	3,341
		12000-13000'	8,048	9,763	3,775 (SNODAS)	10,606 (CU_SWE)	879	1,096
		13000-14000'	347	381	122 (SNODAS)	538 (CU_SWE)	41	42
	San Luis	7000-8000'	4,052	1,493	1,480 (CU_SWE)	9,183 (UA_SWE)	1,789	-536
		8000-9000'	3,208	2,726	2,680 (UA_SWE)	4,217 (CU_SWE)	628	760
		9000-10000'	5,939	7,337	2,870 (CU_SWE)	7,611 (SNODAS)	1,180	2,341
		10000-11000'	13,707	16,741	6,089 (CU_SWE)	18,292 (SNODAS)	1,892	2,017
		11000-12000'	18,093	18,539	16,533 (CU_SWE)	19,206 (SNODAS)	1,783	2,162
		12000-13000'	13,294	13,799	10,094 (SNODAS)	15,989 (CU_SWE)	1,135	1,340
		13000-14000'	303	339	120 (SNODAS)	450 (CU_SWE)	29	36
	Saguache	7000-8000'	910	372	138 (SNODAS)	2,219 (UA_SWE)	728	353
		8000-9000'	3,766	959	577 (CU_SWE)	9,763 (UA_SWE)	924	81
		9000-10000'	12,956	11,574	3,833 (CU_SWE)	23,461 (UA_SWE)	1,259	3,852
		10000-11000'	24,102	28,555	14,713 (CU_SWE)	29,040 (UA_SWE)	2,400	4,654
		11000-12000'	17,208	15,321	14,426 (SNODAS)	21,879 (CU_SWE)	1,369	1,366
		12000-13000'	4,215	4,077	2,499 (SNODAS)	6,069 (CU_SWE)	331	354
		13000-14000'	174	117	32 (SNODAS)	372 (UA_SWE)	10	4
	Conejos	7000-8000'	95	0	0 (SNODAS, CU_SWE)	284 (UA_SWE)	-67	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-08 (af)	Median Vol. Change since 2026-02-08 (af)
Rio Grande Headwaters	Conejos	8000-9000'	1,950	2,820	113 (CU_SWE)	2,917 (UA_SWE)	-655	-481
		9000-10000'	25,245	26,719	17,524 (CU_SWE)	31,492 (SNODAS)	818	2,070
		10000-11000'	46,181	42,673	38,049 (CU_SWE)	57,819 (SNODAS)	4,021	3,924
		11000-12000'	30,555	31,102	29,273 (SNODAS)	31,292 (UA_SWE)	2,606	2,651
		12000-13000'	3,009	3,234	1,852 (SNODAS)	3,940 (UA_SWE)	252	166

Upper Rio Grande

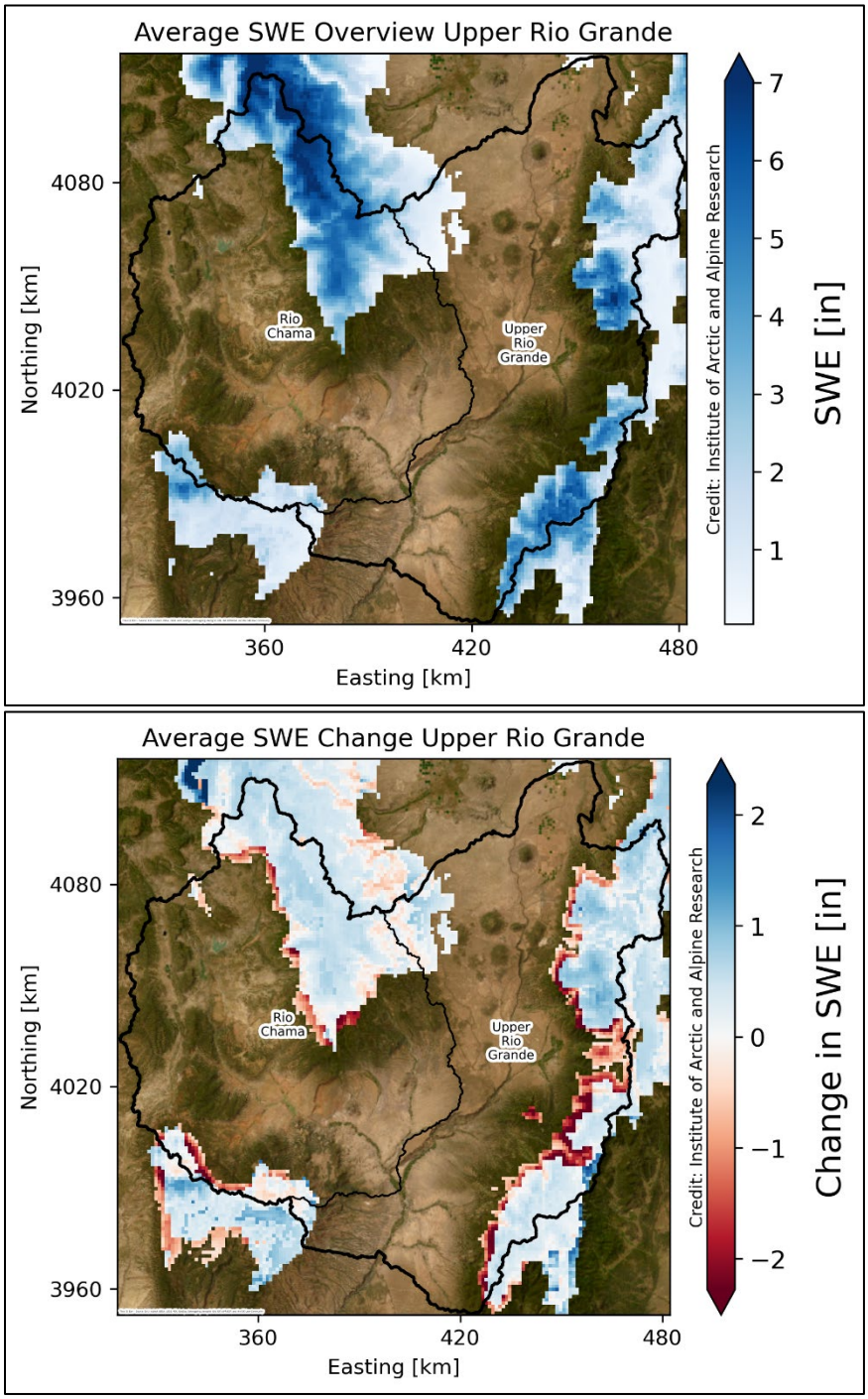

Figure 4. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Upper Rio Grande HUC6 Basin for February 15, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Upper Rio Grande Basin, showing spatial patterns of SWE across subbasins including Rio Chama and Upper Rio Grande (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Upper Rio Grande Basin from February 8, 2026, to February 15, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

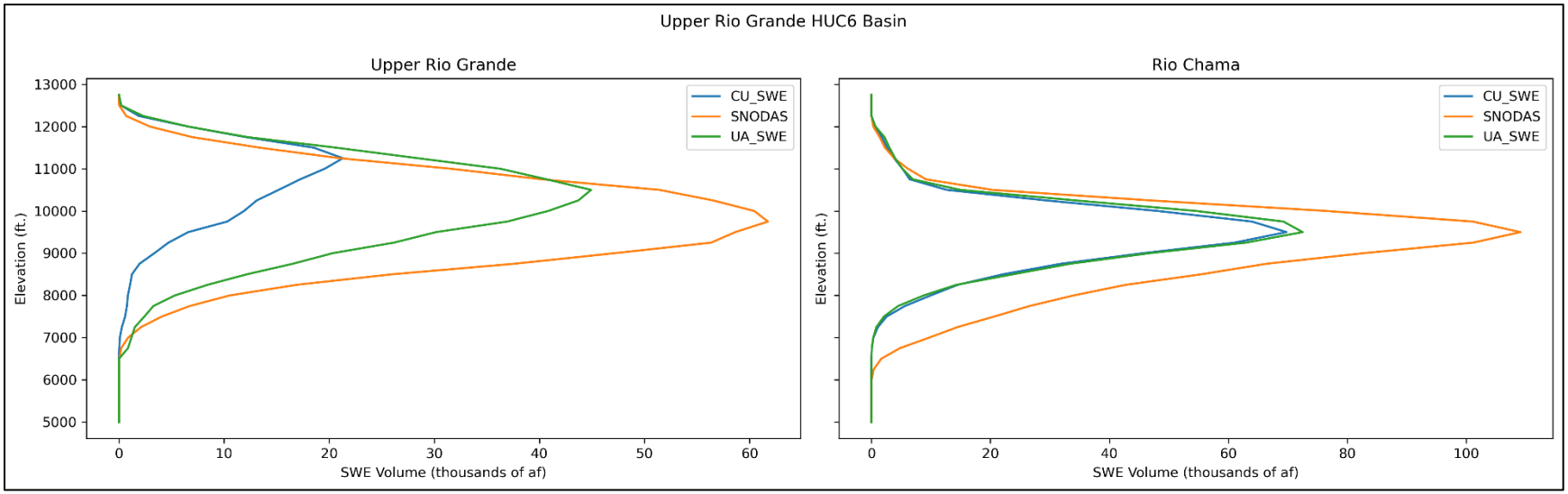

Figure 5. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Upper Rio Grande HUC6 Basin for February 15, 2026. *Elevation-distributed SWE volume for the Upper Rio Grande HUC6 Basin and associated subbasins (Rio Chama and Upper Rio Grande), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 4. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Upper Rio Grande HUC6 Basin for February 15, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 8, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-08 (af)	Median Vol. Change since 2026-02-08 (af)
Upper Rio Grande	Upper Rio Grande	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	711	867	82 (CU_SWE)	1,183 (UA_SWE)	-2,094	-655
		8000-9000'	5,568	5,320	846 (CU_SWE)	10,538 (SNODAS)	-1,373	-1,453
		9000-10000'	23,608	20,323	3,361 (CU_SWE)	47,140 (SNODAS)	582	-3,873
		10000-11000'	37,719	40,824	11,895 (CU_SWE)	60,439 (SNODAS)	5,242	4,722
		11000-12000'	29,155	31,602	19,569 (CU_SWE)	36,293 (UA_SWE)	3,398	6,131
		12000-13000'	5,367	6,580	2,933 (SNODAS)	6,587 (UA_SWE)	520	782
	Rio Chama	5000-6000'	0	0	0	0	0	0
		6000-7000'	8	0	0 (UA_SWE, CU_SWE)	24 (SNODAS)	8	0
		7000-8000'	3,439	333	296 (UA_SWE)	9,690 (SNODAS)	231	-929
		8000-9000'	17,580	9,959	8,785 (UA_SWE)	33,995 (SNODAS)	590	-5,838
		9000-10000'	58,335	46,739	45,593 (CU_SWE)	82,673 (SNODAS)	4,396	424
		10000-11000'	59,791	54,633	48,169 (CU_SWE)	76,572 (SNODAS)	5,126	4,819
		11000-12000'	5,515	5,214	5,199 (UA_SWE)	6,131 (SNODAS)	440	287
		12000-13000'	466	426	297 (SNODAS)	676 (UA_SWE)	33	20

Rio Grande-Elephant Butte (Middle Rio Grande)

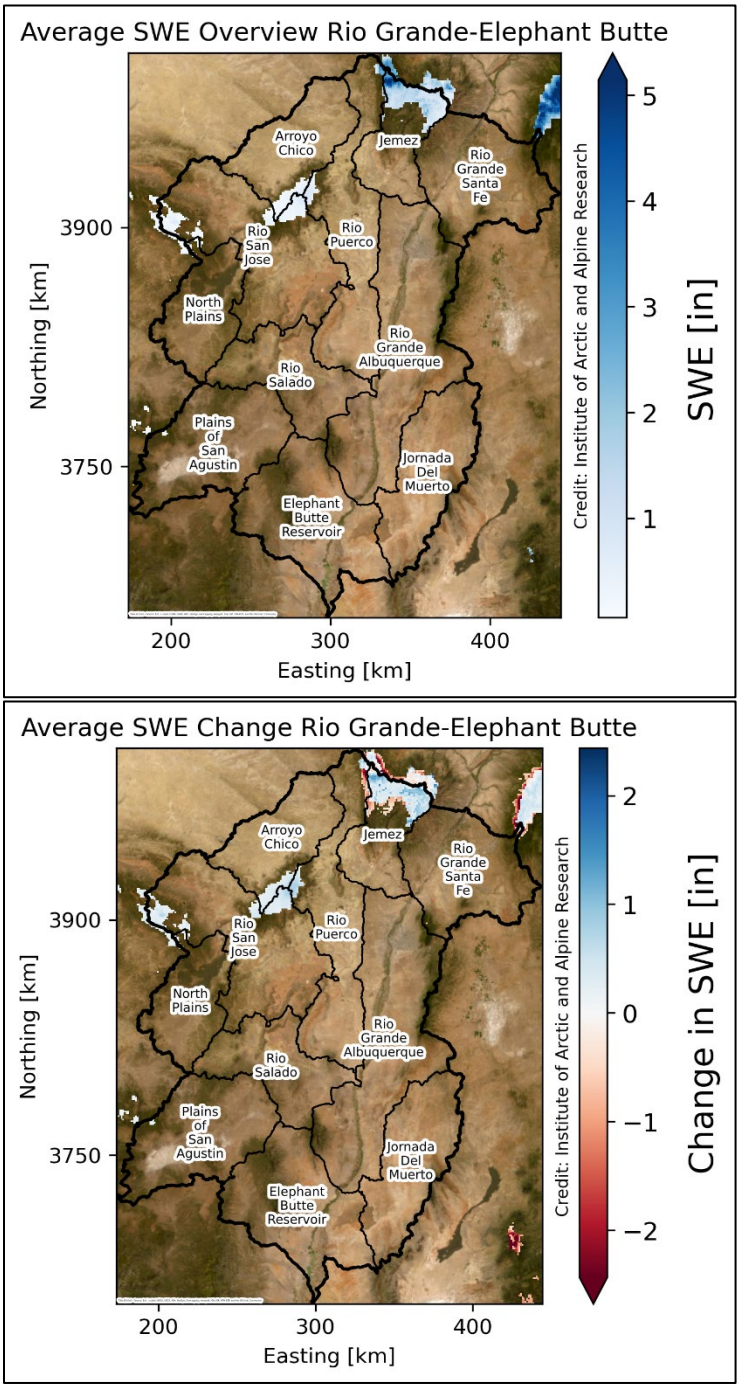

Figure 6. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande-Elephant Butte HUC6 Basin for February 15, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Rio Grande-Elephant Butte Basin, showing spatial patterns of SWE across subbasins including Rio Grande-Santa Fe, Jemez, Arroyo Chico, Rio San Jose, Rio Puerco, Rio Grande-Albuquerque, Rio Salado, North Plains, Plains of San Agustin, Elephant Butte Reservoir, and Jornada Del Muerto (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande-Elephant Butte Basin from February 8, 2026, to February 15, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

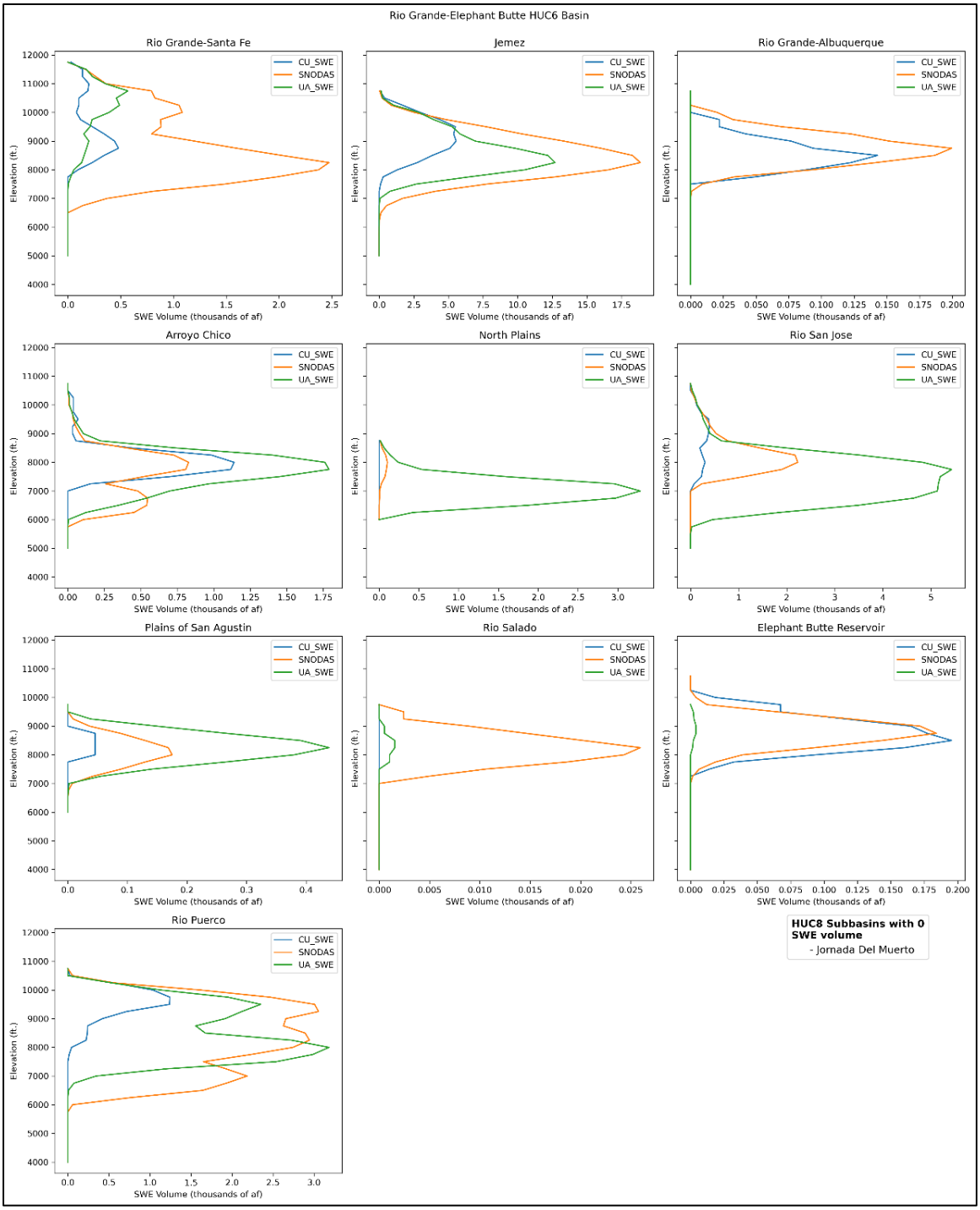

Figure 7. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Rio Grande-Elephant Butte HUC6 Basin for February 15, 2026. *Elevation-distributed SWE volume for the Rio Grande-Elephant Butte HUC6 Basin and associated subbasins (Rio Grande-Santa Fe, Jemez, Arroyo Chico, Rio San Jose, Rio Puerco, Rio Grande-Albuquerque, Rio Salado, North Plains, Plains of San Agustin, Elephant Butte Reservoir, and Jornada Del Muerto), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 5. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Rio Grande-Elephant Butte HUC6 Basin for February 15, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 8, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-08 (af)	Median Vol. Change since 2026-02-08 (af)
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	124	0	0 (UA_SWE, CU_SWE)	371 (SNODAS)	115	0
		8000-9000'	842	96	57 (UA_SWE)	2,374 (SNODAS)	629	-110
		9000-10000'	613	442	199 (UA_SWE)	1,198 (SNODAS)	19	-164
		10000-11000'	518	391	81 (CU_SWE)	1,082 (SNODAS)	37	-10
		11000-12000'	305	356	200 (CU_SWE)	361 (SNODAS)	28	62
	Jemez	5000-6000'	0	0	0	0	0	0
		6000-7000'	1	0	0 (UA_SWE, CU_SWE)	2 (SNODAS)	1	0
		7000-8000'	577	50	0 (CU_SWE)	1,680 (SNODAS)	396	29
		8000-9000'	9,473	10,530	1,314 (CU_SWE)	16,575 (SNODAS)	2,724	6,707
		9000-10000'	8,609	6,932	5,558 (CU_SWE)	13,338 (SNODAS)	1,677	224
		10000-11000'	2,780	2,858	2,522 (SNODAS)	2,959 (CU_SWE)	70	87
	Rio Grande-Albuquerque	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	-1	0
		8000-9000'	60	88	0 (UA_SWE)	93 (SNODAS)	-17	-2
		9000-10000'	76	77	0 (UA_SWE)	152 (SNODAS)	-5	32
		10000-11000'	7	0	0 (UA_SWE, CU_SWE)	20 (SNODAS)	-17	-1
	Arroyo Chico	5000-6000'	0	0	0	0	0	0
		6000-7000'	36	3	0 (CU_SWE)	105 (SNODAS)	36	3
		7000-8000'	392	479	0 (CU_SWE)	698 (UA_SWE)	361	479
		8000-9000'	1,243	1,140	828 (SNODAS)	1,762 (UA_SWE)	-51	1,140
		9000-10000'	73	81	33 (CU_SWE)	106 (UA_SWE)	55	70
		10000-11000'	19	11	7 (SNODAS)	38 (CU_SWE)	-4	-3
	North Plains	6000-7000'	0	0	0	0	0	0
		7000-8000'	1,096	5	0 (CU_SWE)	3,283 (UA_SWE)	1,096	5
		8000-9000'	114	104	0 (CU_SWE)	240 (UA_SWE)	114	104

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-08 (af)	Median Vol. Change since 2026-02-08 (af)
Rio Grande-Elephant Butte	Rio San Jose	5000-6000'	0	0	0	0	0	0
		6000-7000'	152	0	0 (SNODAS, CU_SWE)	455 (UA_SWE)	152	0
		7000-8000'	1,712	3	0 (CU_SWE)	5,131 (UA_SWE)	1,712	3
		8000-9000'	2,451	2,229	301 (CU_SWE)	4,824 (UA_SWE)	1,848	2,079
		9000-10000'	438	408	372 (CU_SWE)	533 (SNODAS)	282	374
		10000-11000'	136	140	127 (CU_SWE)	141 (UA_SWE)	17	39
	Plains of San Agustin	6000-7000'	0	0	0	0	0	0
		7000-8000'	3	2	0 (CU_SWE)	7 (SNODAS)	3	2
		8000-9000'	200	175	46 (CU_SWE)	380 (UA_SWE)	150	165
		9000-10000'	62	36	0 (CU_SWE)	149 (UA_SWE)	51	25
	Rio Salado	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	8	1	0 (CU_SWE)	24 (SNODAS)	8	1
		9000-10000'	3	1	0 (CU_SWE)	9 (SNODAS)	3	1
	Jornada Del Muerto	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
	Elephant Butte Reservoir	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	44	39	0 (UA_SWE)	93 (CU_SWE)	-4	39
		9000-10000'	114	165	4 (UA_SWE)	172 (SNODAS)	-21	163
		10000-11000'	11	11	4 (SNODAS)	19 (CU_SWE)	3	3
	Rio Puerco	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	19	0	0 (UA_SWE, CU_SWE)	58 (SNODAS)	19	0
		7000-8000'	843	343	0 (CU_SWE)	2,185 (SNODAS)	-123	221

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-08 (af)	Median Vol. Change since 2026-02-08 (af)
		8000-9000'	1,990	2,743	46 (CU_SWE)	3,182 (UA_SWE)	1,083	2,294
Rio Grande-Elephant Butte	Rio Puerco	9000-10000'	1,664	1,916	420 (CU_SWE)	2,656 (SNODAS)	372	719
		10000-11000'	1,257	1,126	1,035 (CU_SWE)	1,612 (SNODAS)	60	4

About SWEET

The following information provides a brief overview of SWEET, and the SWE products currently included in the ensemble framework. For more detailed descriptions, citations, and additional information, please refer to the GitHub repository.

SWE products currently used include:

1. **SNODAS SWE**
    a. The SNOw Data Assimilation System (SNODAS) is a physically based snow model developed by NOAA's National Operational Hydrologic Remote Sensing Center and operational since 2004. SNODAS produces daily 1-km gridded estimates of SWE and related snow variables by combining modeled weather inputs with satellite, airborne, and in situ snow observations.
2. **UA/SWANN SWE**
    a. The University of Arizona/Snow Water Artificial Neural Network Modeling System (UA/SWANN) is a research SWE product that uses snow models, artificial neural networks, and assimilated in situ measurements to generate daily gridded SWE and snow depth. The dataset provides 4-km and downscaled 800-m SWE across the conterminous United States, incorporating SNOTEL and COOP observations with PRISM precipitation and temperature fields.
3. **CU SWE**
    a. The University of Colorado (CU) SWE-Fusion product is an operational dataset produced by INSTAAR that provides near–real-time SWE estimates at 500-m resolution. The approach fuses SNOTEL observations with historical SWE reanalyses and terrain variables using a generalized linear model, and scales results using satellite-derived fractional snow-covered area from Snow Today.

The following notes describe key processing methods, assumptions, and visualization conventions used in generating SWE Evaluation Tool figures, tables, and elevation plots.

Figures:

- For ensemble figures, CU SWE and UA SWE are reprojected to the SNODAS resolution (1 sq km) and averages are only taken at pixels where at least 2 of the 3 products have SWE values greater than 0 inches.
- All overview and subbasin plots set pixels with values of 0 inches to NA so that the basemap is visible.
- For individual product figures, each product is displayed at its native resolution.
- Elevation bands of 250 feet are used in the elevation figures, and SWE volume is calculated for each product's native resolution in each band.
- Difference plots display pixels where snow was present in either the current or previous report.
- Figures are available in PNG format on the GitHub repository.

Tables:

- For all tables, mean, median, and volume are calculated at each product's native resolution - zeros are not removed from mean and median calculations.
- For ensemble tables, mean and median volume are calculated using the total volumes from the 3 products - not from an average aggregate raster.
- For min. and max., if 2 products have 0 volume, both products are listed in that cell. If all 3 products have 0 volume, no product is listed for min. or max. in that row.
- Change in SWE will only be calculated starting with the second report of the water year.
- Ensemble calculations: For HUC8 subbasins, the ensemble (mean and median) SWE and volume change (mean and median) are calculated individually for each subbasin. For the HUC6 "Total" row (tables 1 and 2), the ensemble is recalculated using the full HUC6 basin extent rather than summing HUC8. While the ensemble mean is additive and therefore consistent between HUC8 sums and the HUC6 total, the median reflects the middle value among the three products for the entire basin. As a result, median values will not sum to the HUC6 total, since they represent the difference between basin-wide medians rather than the sum of subbasin medians.
- Tables are available in CSV and PDF format on the GitHub repository.
- Elevation tables presented in the report (Tables 3, 4, and 5) are combined into one table (Table 3) on the GitHub repository.

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